

Business Report

Telecom Customer Experience Analysis

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1. Business Understanding

1.1 Business Problem Statement

The telecom company is experiencing inconsistent customer satisfaction due to variations in network performance across operators, regions, and usage environments. Management requires data-driven insights to identify performance gaps and improve service quality.

1.2 Business Objective

Analyse telecom customer experience data to evaluate network performance, identify the key factors influencing customer satisfaction, and provide actionable recommendations for service improvement.

1.3 Business Questions

- Which operator is best rated in each state overall and across indoor, outdoor, and travelling conditions?
 - Is there a relationship between network type and customer ratings?
 - Does call drop have an impact on customer ratings?
 - Where do call drops occur more frequently—indoors or outdoors—and is the difference statistically significant?
 - How can key business metrics be effectively monitored through an interactive dashboard?
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1.4 Business Constraints

- The analysis is limited to the available historical customer experience dataset.
 - The project does not include real-time network performance monitoring.
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1.5 Business Success Criteria

- Identify the key factors affecting customer satisfaction.
- Provide actionable insights to support network optimization.
- Enable data-driven decision-making through an interactive dashboard.

2. Dataset Overview

2.1 Dataset Description

The dataset used in this project contains customer feedback on voice call quality collected through the **TRAI MyCall** mobile application. It includes customer ratings along with network-related information such as telecom operator, network type, customer location, and state. The dataset, published by the **Telecom Regulatory Authority of India (TRAI)**, is used to analyse customer satisfaction and evaluate telecom service quality across different operators and regions.

2.2 Dataset Source

Attribute	Details
Dataset Name	Indian Voice Call Quality Customer Experience Dataset
Source	Telecom Regulatory Authority of India (TRAI)
Data Collection Method	Customer feedback collected through the TRAI MyCall application
File Format	CSV
Data Type	Structured Data

2.3 Dataset Summary

Metric	Value
Total Records	5098
Total Features	8
Telecom Operators	Airtel, BSNL, RJio, VI
Network Types	2G, 3G, 4G, Unknown
Customer Locations	Indoor, Outdoor, Travelling
Call Categories	Satisfactory, Poor Voice Quality, Call Dropped

2.4 Data Dictionary

Column Name	Description
operator	Telecom operator used by the customer.
inout_travelling	Customer location during the call (Indoor, Outdoor, Travelling).
network_type	Network technology used during the call (2G, 3G, 4G, Unknown).
rating	Customer satisfaction rating for voice call quality.
calldrop_category	Customer-reported call quality category.
latitude	Latitude of the customer when the feedback was recorded.
longitude	Longitude of the customer when the feedback was recorded.
state_name	State from which the customer submitted the feedback.

2.5 Dataset Characteristics

- The dataset consists of customer feedback collected from telecom users across multiple states in India.
- Each record represents the voice call experience reported by an individual customer.
- The dataset contains both **categorical** and **numerical** attributes.
- Geographic coordinates enable location-based analysis and map visualisation.
- The dataset supports analysis of customer satisfaction, network performance, and telecom operator comparisons.

3. Data Preparation

3.1 Data Loading

The dataset consisted of multiple monthly CSV files. All files were imported and combined into a single dataset. A **Month** column was created by extracting the month from each file name to support month-wise analysis. A separate working copy was maintained for preprocessing to preserve the original data.

3.2 Initial Data Exploration

The dataset was examined to understand its structure, data types, missing values, and feature distributions. This assessment helped identify data quality issues and determine the required preprocessing steps.

3.3 Data Cleaning

An invalid entry ("**Unnamed: 7**") in the **State Name** column was identified and replaced with a null value (NaN) to accurately represent missing information.

3.4 Missing Values and Duplicate Records

- **Missing Values:** The **State Name** column contained **1,641** missing values, with corresponding latitude and longitude values recorded as **-1**. These records were retained as other attributes remained valid.
 - **Duplicate Records:** A total of **1,979** potential duplicate records were identified. As the dataset lacked a unique identifier, these records were retained to avoid removing valid observations.
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3.5 Data Transformation

- Added a **Month** column by extracting the month from each source file.
 - Consolidated all monthly datasets into a single dataset.
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3.6 Data Validation

The processed dataset was validated by verifying data types, checking data consistency and ensuring it was ready for exploratory analysis, statistical testing.

4. Exploratory Data Analysis (EDA)

4.1 Network Type Distribution by Operator

Objective: Analyse the distribution of network types across telecom operators.

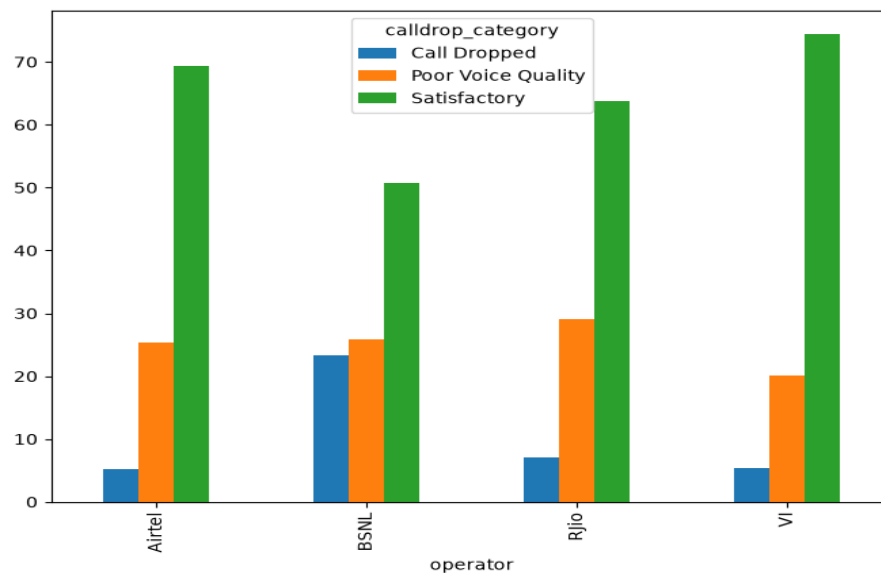
[5]: network_type	2G	3G	4G	Unknown
operator				
Airtel	14.784779	0.124766	54.897068	30.193387
BSNL	1.036269	70.466321	5.699482	22.797927
RJio	0.000000	0.074212	88.014842	11.910946
VI	0.823723	1.153213	64.909390	33.113674

Observations:

- Over 50% of Airtel and VI users were connected to the 4G network.
 - Nearly 90% of RJio users used the 4G network.
 - Most BSNL users were connected to the 3G network.
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4.2 Call Quality Analysis by Operator

Objective: Compare call quality across telecom operators.

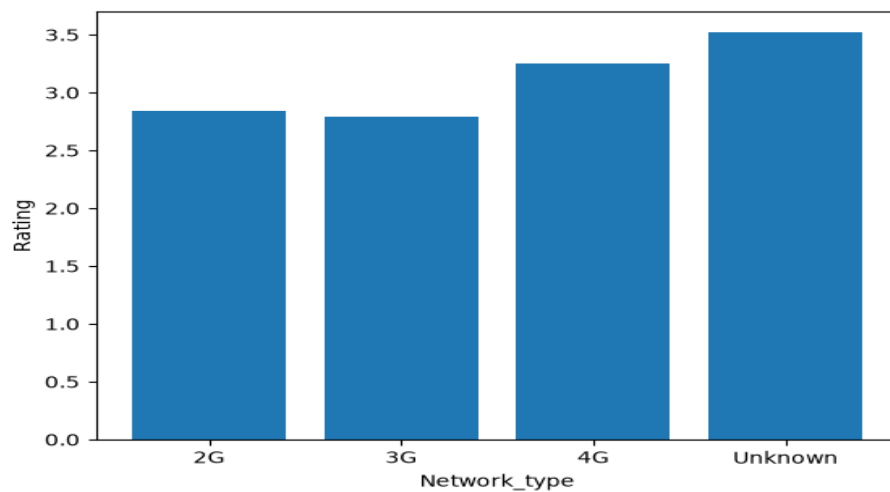


Observations:

- VI recorded the highest proportion of satisfactory calls.
- Airtel maintained a low percentage of call drops.
- BSNL experienced the highest proportion of call drops.
- RJio reported a comparatively higher percentage of poor voice quality.

4.3 Network Type vs Customer Rating

Objective: Examine the relationship between network type and customer ratings.

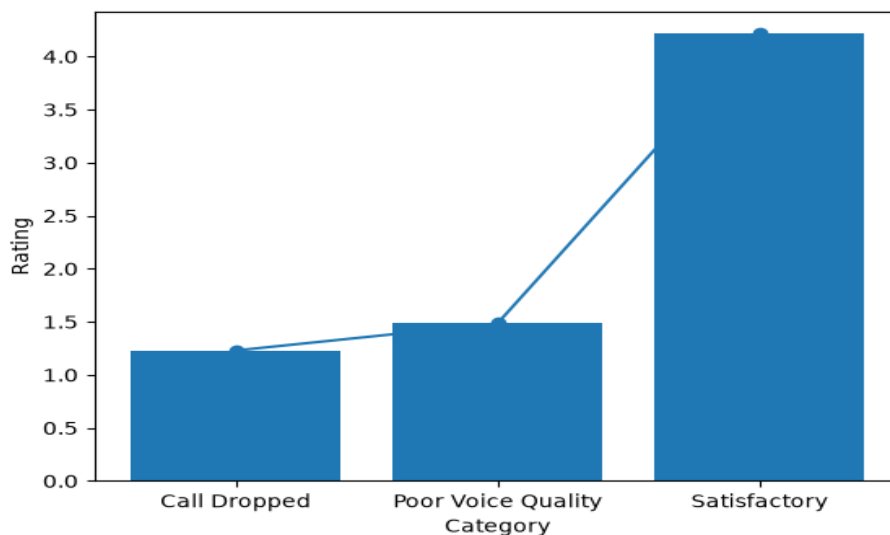


Observations:

- Customer ratings generally increased with newer network technologies.
 - The **Unknown** network category also showed relatively high ratings; however, its actual network type could not be determined from the available data.
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4.4 Call Drop vs Customer Rating

Objective: Analyse the impact of call drops on customer ratings.

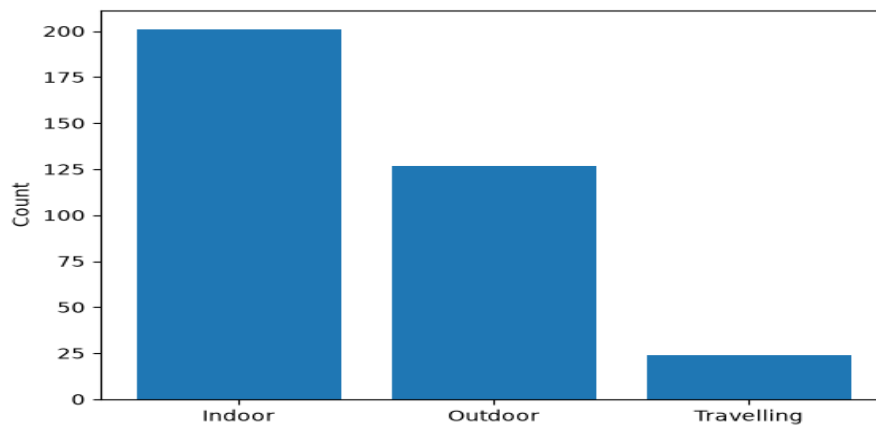


Observations:

- Records classified as **Call Dropped** consistently received lower customer ratings than satisfactory calls.

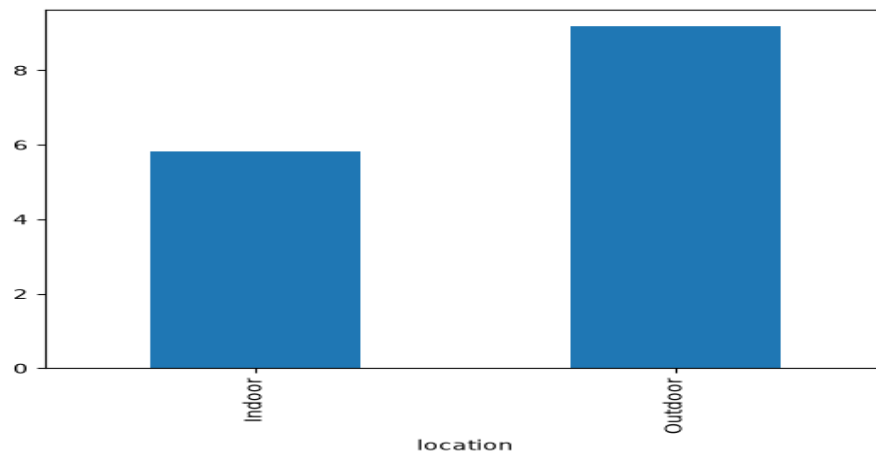
4.5 Location-wise Call Drop Analysis

Objective: Compare call drops across customer locations.



Transformation:

- The **Travelling** category was combined with **Outdoor** to create a new **Location** feature, enabling a direct comparison between **Indoor** and **Outdoor** customer experiences.



Observations

- Based on the raw count of dropped calls, **indoor locations** reported the highest number of call drops.
- To enable a fair comparison, the **Travelling** category was combined with **Outdoor**, and the analysis was repeated using percentage distribution.
- The percentage analysis showed that **outdoor locations** had a higher proportion of call drops than indoor locations.
- This suggests a stronger association between **outdoor usage** and **call drops**.

5. Statistical Analysis

5.1 Relationship Between Network Type and Customer Rating (ANOVA)

Objective

To determine whether customer ratings differ significantly across different network types.

Exploratory Findings

Exploratory analysis indicated that customer ratings generally increased with newer network technologies (2G to 4G). The **Unknown** network category also showed relatively high ratings; however, there is no evidence to confirm its actual network type. Therefore, it is interpreted as an unidentified network category rather than being assumed to represent 5G.

Hypotheses

- **Null Hypothesis (H_0):** The average customer rating is the same across all network types.
- **Alternative Hypothesis (H_1):** At least one network type has a different average customer rating.

Test Performed

A **One-Way ANOVA** test was conducted to compare the mean customer ratings across the four network types (2G, 3G, 4G, and Unknown).

Results

Metric	Value
F-statistic	20.406
p-value	3.85×10^{-13}

Conclusion

Since the **p-value (3.85×10^{-13})** is less than the significance level of **0.05**, the null hypothesis is rejected. This indicates a **statistically significant difference** in average customer ratings across network types. Therefore, network type has a significant influence on customer ratings.

5.2 Association Between Location and Call Drop Category (Chi-Square Test)

Objective

To determine whether there is a significant association between customer location (Indoor/Outdoor) and call drop category.

Exploratory Findings

The percentage-based analysis indicated that outdoor locations experienced a higher proportion of call drops than indoor locations. A Chi-Square test was conducted to determine whether this observed association was statistically significant.

Hypotheses

- **Null Hypothesis (H_0):** There is no association between customer location and call drop category.
- **Alternative Hypothesis (H_1):** There is an association between customer location and call drop category.

Test Performed

A **Chi-Square Test of Independence** was conducted using the Indoor and Outdoor location categories and the corresponding call drop categories.

Results

Metric	Value
p-value	5.27×10^{-5}

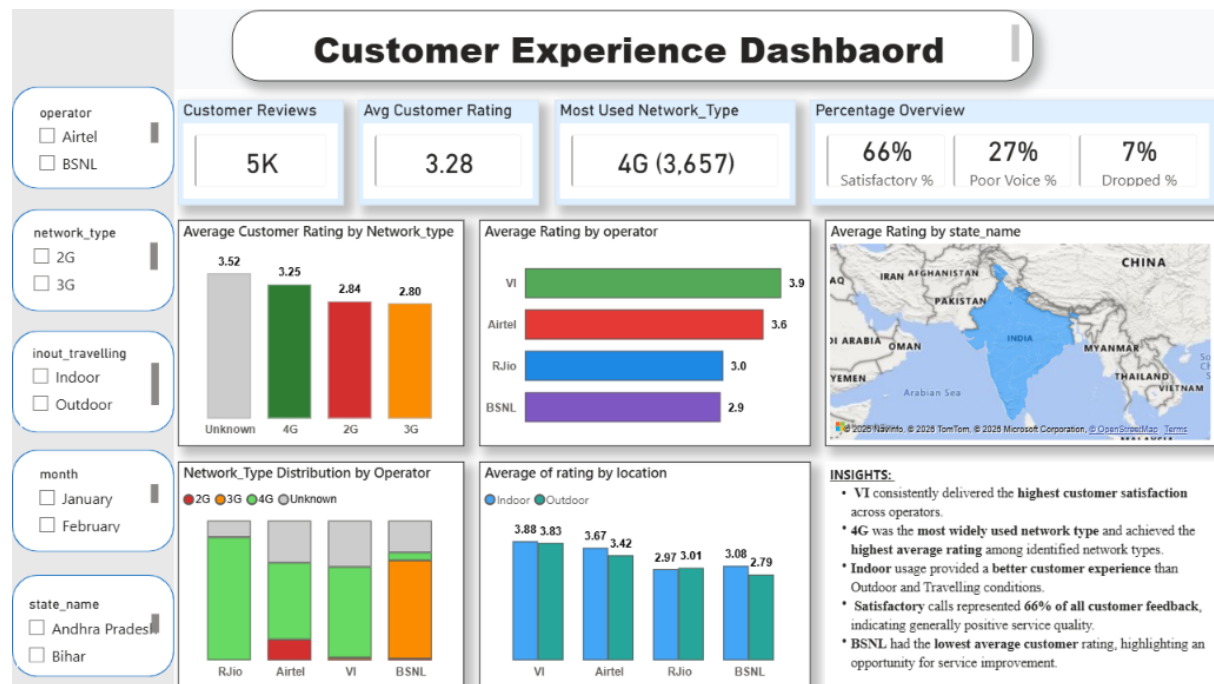
Conclusion

Since the **p-value (5.27×10^{-5})** is less than the significance level of **0.05**, the null hypothesis is rejected. This indicates a **statistically significant association** between customer location (Indoor/Outdoor) and call drop category. Therefore, customer location has a significant influence on the likelihood of experiencing different call quality outcomes.

6. Dashboard Development

6.1 Dashboard Objective

An interactive Power BI dashboard was developed to monitor key customer experience metrics, compare telecom operator performance, and provide actionable insights for business decision-making.



6.2 Dashboard Components

Visual	Purpose
KPI Cards	Display key business metrics such as Average Rating, Satisfactory %, Call Dropped %, Poor Voice %, Total Feedback, and Most Used Network Type.
Operator-wise Rating	Compare average customer ratings across telecom operators.
Network Type Analysis	Analyse customer ratings by network technology.
State-wise Rating Map	Visualise regional differences in customer satisfaction.
Network Type Distribution	Show the distribution of 2G,3G,4G and Unknown in each Operator.
Indoor vs Outdoor Analysis	Compare customer experience across locations.
Slicers	Enable filtering by Month, State, Operator, Network Type, and Location.

7. Key Findings

- **VI** recorded the **highest average customer rating (3.88)**, while **BSNL** recorded the **lowest (2.94)**.
 - Customer ratings differed significantly across network types, with **4G users** generally reporting **higher ratings** (ANOVA, $p < 0.05$).
 - **Call drops** were associated with **lower customer ratings**, indicating a negative impact on customer satisfaction.
 - **Outdoor locations** exhibited a **higher proportion of call drops** than indoor locations, and the **association between location and call drop category was statistically significant** (Chi-Square, $p < 0.05$).
 - Approximately **66% of customer feedback** was classified as **satisfactory**, indicating an overall positive customer experience.
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8. Business Recommendations

- Improve network reliability in outdoor locations to reduce call drops.
 - Prioritize network optimization for lower-performing operators, particularly BSNL.
 - Expand and maintain 4G infrastructure to enhance customer experience.
 - Continuously monitor customer feedback through interactive dashboards to support data-driven decision-making.
 - Perform regular state-wise performance analysis to identify regions requiring targeted improvements.
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9. Conclusion

This project analysed telecom customer experience data using **Python, SQL, and Power BI** to evaluate operator performance, network quality, and customer satisfaction. The analysis identified significant relationships between **network type, call quality, customer location, and ratings**, which were validated through statistical testing. An interactive dashboard was developed to support continuous monitoring and enable informed business decisions for improving telecom service quality.